Trøndelagkonkurranse 2026 - Level II

Third Round - November 4, 2026

2nd Part

Frida wrote the following sequence of numbers

$$011000111100000111111\ldots$$

starting with 0, then two 1's, then three 0's, then four 1's and so on. Skye then asked Frida: 'How many numbers are you going to write?'. To which she replied, 'Just 2026 numbers'.

Task 1: Determine if the number in the 2026th position is a 0 or a 1.

Skye, thrilled, started doing something similar, but instead he wrote the following sequence

$$112123123412345123456\ldots$$

each time concatenating the natural numbers from 1 to n and then continuing with 1 to $n+1$. Frida then asked Skye: 'How many numbers are you going to write?'. To which he replied, 'Just 2026 numbers'.

Task 2: Determine which the number is in the 2026th position.

Peter, who came late to the meeting, saw what they were doing and also wanted to share a problem with them: either using numbers beyond the naturals (like negatives or rationals), numbers arranged in a different way other than a line, or anything interesting he can think of.

Task 3: Help Peter create a nice problem he can give to Frida and Skye.

Task 1:

Let's write the position of the last 0 or 1 of the first few blocks of 0's and 1's

$$\begin{array}{cccccccccccccccccccccc} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & & 3 & & 6 & & & & 10 & & & & 15 & & & & & & 21 & \end{array}$$

So by construction, these values follow the sequence

$$1, 3, 6, 10, 15, 21, \ldots, \frac{n(n+1)}{2}$$

for $n \geq 1$, also known as the triangular numbers. Now, for $n = 1$ we start with a 0, then we alternate between 0 and 1, which means that each time n is odd, the number in the position $n(n+1)/2$ is a 0, and if n is even, the number in the position $n(n+1)/2$ is a 1.

Observe now that for $n = 63$,

$$\frac{n(n+1)}{2} = \frac{63(63)}{2} = 63 \cdot 32 = 2016$$

and for $n = 64$,

$$\frac{n(n+1)}{2} = \frac{64(65)}{2} = 32 \cdot 65 = 2080,$$

which means that the number in the 2016th position is a zero; then, from the 2017th to the 2080th position we have 1's. Thus, the value in position 2026 is a 1.

Task 2:

Following the argument from the previous task, notice that if the position of the last 0 or 1 of each chunk of 0's and 1's is given by $n(n+1)/2$, then n indicates the length of this chunk, which makes sense since

$$1 + 2 + 3 + 4 + \ldots + n = \frac{n(n+1)}{2}.$$

This means that the value in position 2016 will be a 63, followed by a 1 in position 2017. Thus, since we are 9 steps away from 2016, we have that the number in position 2026 is a 10.